

Towards AI-Driven Human-Machine Co-Teaming for Adaptive and Agile Cyber Security Operation Centers

MASSIMILIANO ALBANESE, Center for Infrastructure Security in the Era of AI, George Mason University, Fairfax, United States

XINMING OU, University of South Florida, Tampa, United States

KEVIN LYBARGER, Department of Information Sciences and Technology, George Mason University, Fairfax, United States

DANIEL LENDE, University of South Florida, Tampa, United States

DMITRY GOLDGOF, University of South Florida, Tampa, United States

FAAYED AL FAISAL, University of South Florida, Tampa, United States

KRITAN BANSTOLA, University of South Florida, Tampa, United States

ARKA GHOSH, Center for Infrastructure Security in the Era of AI, George Mason University, Fairfax, United States

Security Operations Centers (SOCs) face growing challenges in managing cybersecurity threats due to an overwhelming volume of alerts, a shortage of skilled analysts, and poorly integrated tools. Human-AI collaboration offers a promising path to augment the capabilities of analysts while reducing cognitive overload. To this end, we introduce an AI-driven human-machine co-teaming paradigm that leverages large language models (LLMs) to support threat intelligence, alert triage, and incident response workflows. We present a vision in which LLM-based agents learn from human analysts the tacit knowledge embedded in SOC operations, enabling them to progressively improve their performance on operational tasks. Our approach involves close collaboration with SOCs to refine this process and identify replicable patterns where human-AI co-teaming improves operational efficiency. To illustrate the feasibility of this paradigm, we report lessons learned from a preliminary case study conducted in partnership with a real SOC.

CCS Concepts: • **Human-centered computing** → Collaborative and social computing; • **Information systems** → Decision support systems; • **Computing methodologies** → Artificial intelligence; • **Security and privacy** → Intrusion detection systems.

Additional Key Words and Phrases: Cybersecurity, Security Operations Centers, Human-AI Collaboration, Large Language Models

Authors' Contact Information: Massimiliano Albanese, Center for Infrastructure Security in the Era of AI, George Mason University, Fairfax, Virginia, United States; e-mail: malbanes@gmu.edu; Xinming Ou, University of South Florida, Tampa, Florida, United States; e-mail: xou@usf.edu; Kevin Lybarger, Department of Information Sciences and Technology, George Mason University, Fairfax, Virginia, United States; e-mail: klybarger@gmu.edu; Daniel Lende, University of South Florida, Tampa, Florida, United States; e-mail: dlende@usf.edu; Dmitry Goldgof, University of South Florida, Tampa, Florida, United States; e-mail: goldgof@usf.edu; Faayed Al Faisal, University of South Florida, Tampa, Florida, United States; e-mail: faayed@usf.edu; Kritan Banstola, University of South Florida, Tampa, Florida, United States; e-mail: kbanstola@usf.edu; Arka Ghosh, Center for Infrastructure Security in the Era of AI, George Mason University, Fairfax, Virginia, United States; e-mail: aghosh4@gmu.edu.



This work is licensed under a Creative Commons Attribution 4.0 International License.

© 2026 Copyright held by the owner/author(s).

ACM 1557-6051/2026/6-ART

<https://doi.org/10.1145/3819819>

1 Introduction

Security Operations Centers (SOCs) play a critical role in defending organizations against evolving cyber threats. However, SOC analysts face major challenges, including high alert volumes, repetitive tasks, and insufficient automation. Due to the increasing complexity and interconnectivity of modern networks, the challenges of preventing, detecting, and responding to security incidents are increasingly stretching the operational capacity of SOCs¹, placing an overwhelming burden on human analysts. Real-time threat intelligence is crucial, yet the process of collecting, analyzing, and disseminating relevant information is generally time-consuming, as it relies heavily on human analysts to interpret heterogeneous data and make informed decisions. This delay can leave organizations vulnerable to rapidly evolving threats. Additionally, the persistent shortage of skilled cybersecurity professionals limits the ability of organizations to effectively leverage threat intelligence and respond to emerging threats. As these challenges intensify, the demand for more automation in incident response has emerged as a critical focus in cybersecurity research. Leading IT companies are at the forefront of developing tools and orchestration platforms to enhance the efficiency of security incident handling. IBM Resilient, Splunk Phantom, Microsoft Azure Sentinel, Cisco SecureX Orchestration, and Fortinet FortiSOAR are examples of prominent orchestration platforms that support incident response processes. They connect with various security tools and provide a central platform for managing and responding to security incidents. They aim to improve workflows, reduce response times, and increase incident handling efficiency.

Despite significant advances in tooling support, SOC analyst burnout [54] remains a critical issue that was exacerbated by the COVID-19 pandemic [32]. Several factors contribute to this challenge. First, the sheer volume of alerts generated by security tools — most of which are false positives [3] — places a significant burden on analysts. This flood of alerts is often accompanied by uncertainty and incomplete data, making it difficult to prioritize and respond effectively. SOC analysts typically rely on *runbooks* or standard operating procedures (SOPs) to address these alerts. However, the repetitive application of these procedures to predominantly false alarms leads to both fatigue and a lack of fulfillment in the role [54]. Moreover, mainstream security tools often lack the capability to connect important contextual information crucial for comprehending the threat landscape and adapting incident response strategies. Such contextual information may include the specific assets that require protection based on the organization's business needs, the nature of users in the organization's environment, the compliance requirements of the organization, and the political scenario or the economic situation of the organization's country. A human analyst typically considers these contextual factors to decide whether and how to act on an alert. Such reasoning is difficult to capture within traditional algorithmic workflows used to build SOC tools. As a result, currently available SOC tools provide limited support in alleviating these operational pain points [54]. Sundaramurthy *et al.* have conducted a long-term anthropological study of SOCs spanning more than a decade and multiple corporate and higher education SOCs [54–56]. That work showed that, by becoming an anthropologist and embedding within a SOC, a security researcher can extract tacit knowledge within the SOC environment and create tools that (i) readily fit into the analysts' workflow, (ii) precisely address the pain points that result in burnout, and (iii) expand analysts' capability by proactively hunting and responding to threats [56]. This approach can be explained by the notion of tacit knowledge — a concept introduced by Polanyi [47, 48] — which refers to knowledge held by individuals that has not yet been explicitly articulated.

The study in [55] uncovered extensive tacit knowledge among SOC analysts. To access this knowledge, researchers needed to first establish trust with the analysts by embedding themselves in a SOC, performing the same roles as the analysts while observing daily activities through the lens of an anthropologist. This immersive fieldwork, combined with qualitative analysis of fieldnotes, offered deep insights into operational challenges within SOCs that were not always explicitly recognized by the analysts themselves. Consequently, the researchers

¹In the literature, the terms Cyber Security Operation Center (CSOC) and Security Operation Center (SOC) are often used interchangeably, although the latter is more common.

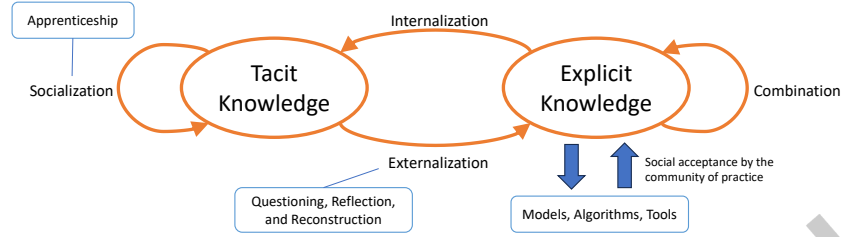


Fig. 1. SECI model for anthropology-guided SOC tool building.

were able to create innovative tools to effectively support the analysts' workflows [54]. This process can be explained by the SECI knowledge model in Fig. 1, first proposed by Nonaka and Takeuchi [45]. Embedded researchers access tacit knowledge by becoming apprentices of SOC analysts and observing their work as it unfolds (*socialization*). Through questioning, reflection, and reconstruction, the researchers externalize the tacit knowledge into explicit forms such as documentation (*externalization*). The accumulated explicit knowledge is combined (*combination*), internalized (*internalization*), and then applied by the apprentices back to their work. This, in turn, forms new tacit knowledge, and the cycle repeats. Working inside the SOC allows researchers to be part of this knowledge conversion cycle, through which tools that are readily accepted by analysts can be developed to codify the explicated knowledge and alleviate the burden on the analysts by automating the most repetitive and time-consuming aspects of the investigation process.

Traditional SOC tools, such as SIEMs, often fail to capture the contextual and tacit knowledge required for effective decision-making. To address these limitations, we propose an AI-driven human-machine co-teaming approach that enhances SOC workflows through adaptive AI agents. This research builds upon our prior anthropological studies of SOC, where researchers embedded within security teams uncovered the importance of tacit knowledge in cybersecurity operations. Here, we extend this work by integrating Large Language Models (LLMs) to serve as AI apprentices, assisting analysts during real-time threat investigations.

To the best of our knowledge, our framework represents one of the first attempts to explicitly capture and operationalize tacit knowledge for improving generative AI agents assisting cybersecurity operations. Our prior work [54–56] investigated the role of tacit knowledge in SOC environments, where anthropological fieldwork was used to inform the design of traditional non-AI tools supporting analysts. In contrast, this work explores how large language models can serve as AI apprentices that learn from analyst interactions and operational context. While AI co-pilots for SOC are beginning to emerge in industry platforms, most current approaches rely primarily on training models on historical datasets rather than enabling structured human-AI co-teaming within operational workflows.

The remainder of the paper is organized as follows. Section 2 presents relevant related work, whereas Section 3 discusses representative incidents that motivate our work. Section 4 introduces our vision and overall approach for human-machine co-teaming in SOC, and Section 5 presents the proposed framework in detail. Section 6 presents a case study and preliminary findings. Section 7 articulates open research challenges and outlines future research directions. Finally, Section 8 concludes the paper.

2 Related Work

While still emerging, the application of LLMs in cybersecurity has been explored in a growing number of task-specific studies, including penetration testing [19, 25, 49], incident response, vulnerability management, policy generation, security training, and detecting threats, malware, and intrusions. Research has primarily explored classification-oriented applications, such as identifying network intrusions and detecting malicious

URLs [16, 26, 38, 50], with some exploration into synthesizing data into unified reports [51], including integrating global threat intelligence with local organizational knowledge [43]. However, research on leveraging LLMs to support SOC analysis and generate actionable outputs, such as creating or refining incident response plans [27], remains relatively limited. While there are several cybersecurity-annotated datasets for training machine learning (ML) models on specific tasks [2, 12, 17, 23, 35], limited work has focused on instruction tuning to distill expert knowledge and enable diverse cybersecurity-specific tasks [37]. Furthermore, the study of continuous learning strategies in this domain remains largely unexplored. Real-world adoption faces challenges, including privacy concerns with proprietary, cloud-based models like ChatGPT, the high computational demands of LLMs, and the need for cybersecurity-specific models that can capture and distill analysts' knowledge into actionable insights aligned with specific environments and the requirements of individual SOC [16, 26, 38, 50]. Additionally, the probabilistic nature of LLMs introduces variability that contrasts with the deterministic tools SOC analysts typically rely on, raising questions about reliability and trust when identical inputs yield different outputs. This issue is compounded by "hallucinations," where LLMs may provide inaccurate or fabricated information, creating concerns about their dependability for SOC operations [16, 26].

We argue that these perceived deficiencies of LLMs are not insurmountable obstacles to their adoption in SOC. Rather, they arise from attempts to use generative AI models as traditional algorithmic tools. By adopting a human-centric perspective and drawing parallels between human SOC analysts and LLMs, we can view an LLM-based AI agent as a collaborator rather than a deterministic tool [25]. A SOC analyst can use the AI agent to elicit ideas on how to proceed with an investigation, and the agent can provide supporting evidence for its suggestions, the validity of which human analysts can verify. This human-machine co-teaming is fundamentally different from how analysts use traditional SOC tools such as SIEMs. Analysts do not blindly trust the output of the AI agent; rather, they evaluate its suggestions and require evidence-based justification. The value provided by an AI agent lies in its ability to process vast amounts of data much faster than a human while navigating the nuanced semantics of that data. When investigating cyber incidents, once the relevant data pieces are presented to a human analyst, it is often quite clear what attackers have done and with what effect. However, identifying those relevant pieces across massive volumes of data can easily overwhelm a human. A machine's upper bound on such bandwidth is much higher than that of the human brain, but it must still be capable of the intricate reasoning required to *connect the dots* within the data. How human analysts connect these dots can rarely be specified through a fixed set of rules; much of this knowledge is tacit and difficult even for analysts themselves to articulate. LLMs therefore present an opportunity to learn such tacit knowledge from data, enabling a form of human-machine co-teaming in which analysts are gradually relieved from the most repetitive tasks, such as processing the large number of alert tickets that ultimately turn out to be false positives [3].

Geoffrey Hinton highlighted parallels between human cognition and AI systems [34]. Like LLMs, human analysts are not immune to inconsistencies, errors, and knowledge gaps. However, SOC analysts mitigate these limitations through rigorous training that blends cybersecurity expertise with organization-specific practices. This training includes explicit knowledge such as regulatory compliance requirements, as well as tacit knowledge gained through experience such as recognizing anomalous patterns in network behavior or interpreting subtle contextual cues from threat intelligence reports. LLMs, like human analysts, can improve their reliability through iterative learning and feedback. By integrating mechanisms to capture both explicit and tacit knowledge, LLMs could evolve into valuable SOC collaborators, supporting analysts in a dynamic threat environment. When working with an LLM-based AI agent, analysts should critically evaluate its suggestions, just as they would evaluate those of a human colleague, requiring evidence-based justification. Viewing an AI agent as a partner rather than a deterministic tool enables SOC teams to utilize its assistance in much the same way they would rely on a human apprentice. Techniques such as prompt engineering, supervised fine-tuning, reinforcement learning from human feedback (RLHF), and retrieval-augmented generation (RAG) can facilitate this adaptation [20, 63]. We envision AI-driven human-machine co-teaming in SOC, where SOC analysts work alongside AI agents,

collaboratively tackling investigation tasks. These agents will enhance analysts' capabilities by internalizing the tacit knowledge required to process nuanced data and help human analysts scale up their work substantially.

In summary, Human-AI collaboration in cybersecurity has been explored in multiple domains, including automated alert triage, threat hunting, and decision support. Existing research on SOC automation highlights the limitations of rule-based systems and the need for adaptive AI-driven approaches. Our work differentiates itself by envisioning a human-centric co-teaming framework, where LLMs learn from real-world SOC workflows through iterative feedback and adaptation.

3 Motivating Examples

In this section, we describe two high-profile incidents that highlight the need for novel AI-driven solutions to support the work of security analysts. While challenges remain — including trust in AI outputs, explainability, and continuous learning from SOC interactions — human-AI co-teaming offers transformative potential to help mitigate complex and large-scale threats. Section 7 discusses open research challenges and outlines future research directions.

Cyber-attacks targeting high-profile organizations are often driven by geopolitical tensions, exposing critical gaps in SOC, particularly in integrating external threat intelligence and leveraging human-machine collaboration for increased efficiency and adaptive response to evolving threats. Two major incidents — the 2014 Sony Pictures Entertainment attack and the 2016 DNC cyber intrusions — illustrate these challenges.

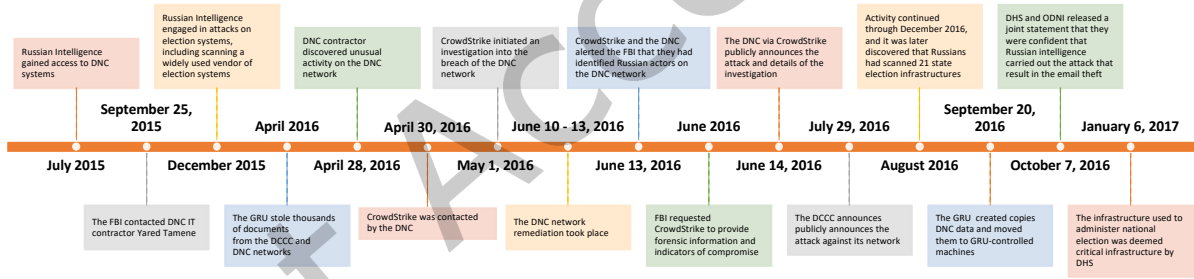


Fig. 2. Timeline of the DNC attack.

The Sony attack, attributed to North Korea [41] in retaliation for the release of the movie *The Interview* [40], began with a phishing campaign that compromised Sony's network. The attackers deployed Destover malware, wiping data and rendering systems inoperable while simultaneously exfiltrating sensitive data, including unreleased films, emails, and employee records. The breach caused severe operational and reputational damage, highlighting the failure of traditional SOC to anticipate politically motivated cyber threats. Similarly, the DNC breaches, orchestrated by Russian state-sponsored groups Cozy Bear and Fancy Bear [18], relied on spear-phishing to gain initial access. Cozy Bear persisted within the network for months before Fancy Bear executed more aggressive data exfiltration and public leaks, influencing U.S. politics (see Fig. 2). These attacks underscored the difficulty of identifying and responding to prolonged, stealthy threats in a complex geopolitical landscape.

Our framework would have significantly enhanced a SOC's ability to detect and mitigate such attacks. By incorporating geopolitical threat signals into SOC workflows, harmonizing disparate data sources, and leveraging LLM-based AI for real-time contextual analysis, our approach would have provided analysts with valuable actionable intelligence well before the attacks escalated. Rather than relying on static detection mechanisms, our adaptive AI-driven co-teaming paradigm would have surfaced geopolitical risks, linked anomalous activity to

external threat actors, and generated explainable remediation plans, allowing human analysts to act swiftly and effectively. The Sony and DNC incidents illustrate the urgent need for a human-machine collaboration model that dynamically integrates external intelligence, anticipates emerging threats, and enables proactive defense.

4 Vision Overview

Our vision for human-machine co-teaming in SOC operations, partly inspired by our anthropological study of SOC, is illustrated in Fig. 3. We propose an LLM-based AI agent that acts as an apprentice to human SOC analysts, assisting through interactions in the execution of subtasks or definition of investigative ideas. The human analyst evaluates the agent's outputs: if the results are inaccurate or unhelpful, the interaction provides additional guidance or refutes the LLM agent's answer with supporting reasoning. A human apprentice would acquire tacit knowledge through such interactions; for an LLM agent, we require a specific learning process to capture these insights and incorporate the knowledge for future use. This learning process ultimately produces an LLM agent specialized to handle SOC tasks effectively.

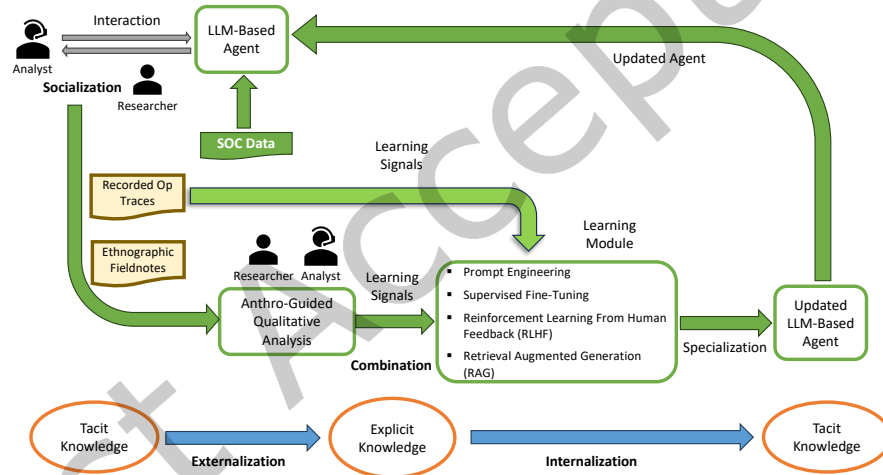


Fig. 3. Our vision for human-machine co-teaming for SOC operations.

The Learning Module utilizes learning signals derived from analyst-LLM interactions. Op traces consist of raw transcripts of conversations between the analyst and the LLM agent. The ethnographic fieldnotes are recorded by a researcher trained in anthropological methods and embedded in the SOC. This researcher observes interactions between the analyst and the LLM, capturing contextual information in fieldnotes that will be analyzed using qualitative research methods to reveal the tacit knowledge hidden in such interactions. Qualitative analysis can extract classifications, assessments, and higher-order concepts that analysts deem important in their work and, through coding, transform this data into annotations [21, 36]. These codes can describe the category or concept, provide relevant examples, and include inclusion and exclusion criteria (all basic parts of qualitative data analysis) [13, 57]. In the initial stages of the learning process, the embedded researcher can also provide a further source of input about gaps between human preferences and AI outputs, which will provide a triangulation between the SOC, the LLM agent, and the research team. Importantly, this triangulation is based on direct observation rather than secondary assessment, thus increasing the validity of what is working and what is missing in ongoing interactions between analysts and the LLM. Op traces, fieldnotes (including scalar descriptions of what is relevant

and non-relevant), and analytic codes from anthropological research can all inform learning signals. These learning signals align the LLM agent's behaviors with the analyst's and the SOC's specific goals.

Given the sensitivity of SOC data, LLMs used in this research must be housed on premises, necessitating the use of open-source pre-trained foundation models (e.g., Falcon [8], LLaMA [58], Mistral [30]). These models are trained on broad and publicly available corpora, encompassing general web data, books, and technical documents, which may include some cybersecurity content; however, they lack the domain-specific expertise and operational context required for effective SOC operations. Our prior study shows that proficiency in SOC operations requires specialized tacit knowledge gained through experience. Thus, our proposed learning process is based on data generated through the AI agent's experience in handling real SOC tasks. This is a critical step in specializing the agent to assist analysts effectively. The agents will *learn on the job*, just as how human SOC analysts do, so their behavior will gradually align with the specific needs of the SOC. In this online learning paradigm, the agents will require less human input to produce useful results. The cycle can be seen as an instantiation of the SECI model (see Fig. 1) in the context of human-LLM co-teaming. It equips the LLM agent with the required tacit knowledge to perform the tasks specific to the SOC, developing a robust human-AI partnership to enhance the effectiveness and efficiency of SOC operations, reduce the cognitive load on analysts, and mitigate burnout.

An important consideration when using LLMs in SOC is whether model bias could hinder the accuracy and utility of the system. On the one hand, human-machine co-teaming hedges against the inherent bias in the model coming from the data it is pre-trained on; on the other hand, the fact that human analyst's interactions with LLM agents become sources of learning signals to shape the AI agent's future behaviors introduces the possibility that the human analyst's bias may be reinforced or amplified by the LLM agents. How to adequately address these potential risks of bias is an important research topic for future study.

Another practical consideration is the legal and ethical framework for using LLMs, particularly those not hosted on premises within the SOC. In our current framework, we assume that the LLMs are based on pre-trained open-source models deployed locally to avoid potential legal and privacy issues associated with transmitting customer data to external entities. However, if our proposed vision proves effective in practice, organizations may eventually seek to leverage more powerful models hosted by specialized "AI factories." This evolution would parallel the adoption of cloud computing, where initial concerns over data security and control led to the emergence of private and community clouds as alternatives to public cloud adoption. Similarly, ensuring strong safeguards for customer data and compliance with applicable legal and regulatory requirements will be essential to enable this next phase in SOC modernization.

4.1 Learning Strategies

Given the constraints that require LLM agents to operate in on-premises environments, it is essential to explore lightweight learning strategies that avoid the intensive resource demands of training new foundation models. Given the rapid development of LLMs and the emergence of new reasoning capabilities, there are many open questions regarding the tasks that these models can perform without any domain- or task-specific training. In this paradigm, where LLMs approach all language processing tasks as generative tasks, a range of learning strategies becomes relevant. Across tasks, our approach begins with prompt engineering, which provides LLM-specific natural language instructions on how to perform a task, serving as the foundational step in all experimentation. By crafting precise prompts, we can evaluate the baseline capabilities of a given LLM, such as Meta's LLaMA, for specific tasks without additional training. This provides critical insight into whether an LLM's inherent capabilities are sufficient or if further refinement is necessary. For tasks where baseline performance proves inadequate, we can employ supervised fine-tuning [42, 63], which involves curating a dataset of input-output pairs tailored to task requirements. This allows the model to specialize in specific tasks by training on the curated dataset. Additionally, we can leverage generative feedback [31], a cutting-edge approach where users provide natural

language critiques of system outputs to guide improvements. This feedback enables users to articulate desired refinements and also provides a mechanism for systematically enhancing the LLM by addressing its current limitations. For more nuanced refinements, particularly in tasks involving subjective evaluation criteria, we can integrate reinforcement learning from human feedback (RLHF) [42, 63], which aligns LLM outputs with human preferences by using feedback such as rankings or scores in fine-tuning. In this process, the researcher ranks multiple system outputs (e.g., A is better than B) to guide the model towards outputs better aligned with human preferences and expert judgment. RLHF enables iterative improvement by embedding human-like preferences into the model's decision-making processes. Finally, when tasks require integrating external or internal knowledge sources, we can utilize retrieval-augmented generation (RAG) [20], which integrates external sources to retrieve relevant resources based on user queries and incorporate them into outputs, grounding responses in a predefined knowledge base and ensuring that outputs are informed by up-to-date and domain-specific data. This layered strategy, progressing from prompt engineering to fine-tuning, RLHF, and RAG, provides a flexible, scalable framework to effectively tackle project subtasks and achieve desired outcomes.

4.2 Overarching Research Questions

Our work is driven by the following two overarching research questions.

Research Question 1. What processes can be designed to enable LLM agents to effectively learn from experience and improve their performance on SOC tasks?

Research Question 2. To what extent can these learning processes enhance LLM agents' performance, resulting in productivity gains that substantially outweigh the effort invested in their development?

Our preliminary research on applying LLM in software security pen-testing [49] shows promise for the first research question, so we will leverage this work in our ongoing efforts. Specifically, after a few iterations of the learning process, we can compare the updated LLM agent's performance against the previous version of the LLM agent on new tasks, utilizing metrics that reflect the agent's helpfulness. One metric could be the number of rounds needed for an analyst to obtain useful information to further the investigation. Our research aims to design specific metrics for an LLM agent, based on the specific tasks it is assigned to work on. For the second research question, we can conduct human subject research to obtain feedback from the SOC analysts and use both qualitative and quantitative assessments to measure the return on investment of adopting the LLM agents.

5 Framework

Fig. 4 provides an overview of our framework for human-machine co-teaming, based on the vision laid out in Section 4. The framework comprises four primary modules, each comprising several submodules.

The framework adopts a data-centric approach that emphasizes cybersecurity integration and interoperability across diverse tools (Module 1). To encourage adoption, the framework complements existing tools and capabilities rather than requiring organizations to overhaul their current systems. By building on established processes, organizations can adopt and integrate the framework with minimal disruption. A key aspect of this approach involves assessing the quality and trustworthiness of data from various sources (Module 1.1). A key innovation is the data normalization mechanism, which leverages LLMs to harmonize data across formats, ensuring tool interoperability and feeding into the next step: context-aware cyber situational awareness (Module 2), which includes an internal graphical data model (Module 2.2). This model is grounded in prior work on attack modeling and configuration security [4, 5] and provides input to an adaptive decision-making process (Module 3), capturing detailed information about security artifacts and their interrelationships. Module 3 utilizes an LLM agent and incorporates data from both Module 1 and Module 2 to analyze and interpret logs, alerts, and network traffic,

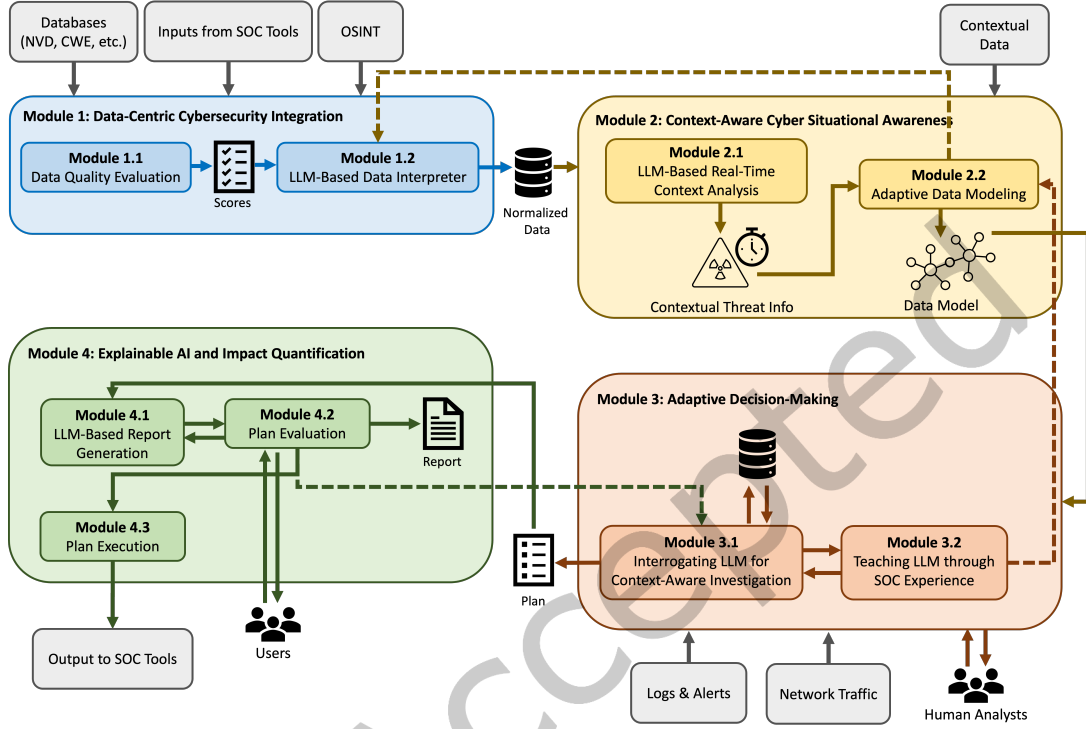


Fig. 4. Overview of the proposed framework.

generating recommended courses of action in response to security events. The LLM agent dynamically adapts to the evolving threat landscape by incorporating real-time context analysis of organizational, economic, and geopolitical data (Module 2.1). This adaptive data modeling improves the accuracy and timeliness of threat assessments by contextualizing security events within their broader environment. Self-awareness mechanisms, based on previous feedback received by analysts in different but analogous situations, enable AI agents to recognize when human expertise is needed — whether to validate findings or contribute to critical decision-making steps. The final plan of action developed through joint human-machine collaboration is passed to the Explainable AI and Impact Quantification module (Module 4). This module utilizes LLMs to generate a human-readable report detailing the recommended course of action (Module 4.1) and its impact on the organization's security posture. The AI-generated explanation is evaluated by various user groups (Module 4.2), and the feedback loops back into the adaptive decision-making process to refine future recommendations. Once validated, the plan is executed by leveraging LLM capabilities from Module 1 to translate it into instructions for the organization's security tools (Module 4.3). The following sections elaborate each module and outline remaining research directions.

5.1 Module 1: Data-Centric Cybersecurity Integration

Module 1 adopts a data-centric approach to address the challenges of integrating a vast and growing array of data sources, platforms, and tools to support interoperability and enable large-scale automation.

Research Questions. How can LLMs be leveraged to address the challenges of integrating diverse cybersecurity tools within SOC, particularly by minimizing configuration complexities, overcoming the inflexibility of existing tools, and ensuring interoperability? Additionally, how can these solutions remain robust against incomplete or uncertain data, as well as evolving data formats and APIs?

Modern SOC handle data in a range of formats used by vulnerability scanners, log analyzers, intrusion detection systems, and external threat intelligence sources. The lack of standardization across these tools presents significant challenges for integration and analysis. This module explores strategies to minimize complexity, thus lowering barriers to adoption, and making the proposed solution robust to incomplete and uncertain data. Given that standardization remains a primary obstacle to integration, we argue that — given the complexity of the problem at hand and the current availability of tools from a large number of vendors — pursuing standardization efforts is not the most effective and efficient way to address the problem, due to several practical challenges like resistance from vendors, dynamic and diverse nature of the cybersecurity domain, regulatory and compliance requirements across industries and regions. To tackle this objective, the framework includes a metrics-based approach for assessing the quality and trustworthiness of data gathered from various sources. An LLM-based mechanism then addresses data interoperability challenges, focusing on the variability in threat and vulnerability report formats and the prevalent use of natural language to describe critical details.

5.1.1 Module 1.1: Evaluation of Data Quality and Trustworthiness. The primary objective of this module is to establish a robust mechanism for evaluating the quality, reliability, and trustworthiness of data ingested from diverse cybersecurity tools and platforms. As SOC increasingly depend on inputs such as network logs, vulnerability scanners, and external threat intelligence, accurately assessing the reliability and completeness of these data sources is essential. Initially, data from various sources is normalized using the mechanisms from Module 1.2. At this stage, the system operates under a neutral trust assumption, treating all sources as equally reliable due to the absence of prior knowledge about their accuracy. Over time, as analysts provide feedback on recommended actions through human-machine cooperation (Module 3) and plan evaluation (Module 4.2), this input is used to infer and adjust the reputation of each data source. A reputation scoring system can be based on a centrality metric approach similar to PageRank [14], where data sources and validated plans are represented as nodes and feedback is encoded as edges. Unlike traditional PageRank, where all edges transfer importance, this approach allows feedback to be positive or negative, reflecting whether a source contributes accurate or misleading data to a plan, and uses a bipartite graph with two classes of nodes — sources and plans. Scores dynamically adjust based on historical performance and analyst feedback, enabling the system to prioritize reliable sources and refine its trust assessments iteratively. This adaptive scoring system allows continuous improvement as real-world outcomes and feedback are integrated, enhancing both data quality and the effectiveness of decision-making within SOC workflows.

5.1.2 Module 1.2: LLM-Based Data Interpretation. An LLM-based agent interprets and normalizes data generated by diverse cybersecurity tools and platforms into a common internal format, translating between formats for seamless interoperability.

Common data format. The diverse data formats include structured, semi-structured, and unstructured data, including natural language descriptions. A JSON format is used to normalize cybersecurity threat and vulnerability data, building on the widely used STIX standard [11]. This format leverages key-value pairs, which are both machine-readable and interpretable by LLMs, enabling the extraction and normalization of diverse data types, including categorical variables (e.g., severity levels, attack vectors) and unstructured text (e.g., threat descriptions). The format integrates data quality and trustworthiness scores from Module 1.1, to propagate these metrics to the Module 2 graph modeling.

Data set curation. To develop and evaluate an LLM agent, the framework curates a parallel corpus of records that span a range of input-output format combinations. This corpus includes input-output pairs created using existing format conversion tools to provide opportunistic data and manually curated samples spanning format conversions not addressed through available tools to broaden the training coverage. Vendor-provided data format specifications are incorporated into the LLM instructions to ground the format conversion. These specifications serve as a knowledge source and update path for the LLM agent as formats change or new formats are released. The curated input-output pairs emphasize syntactic diversity and semantic consistency across heterogeneous data formats. Each input represents a structured record or configuration file conforming to a specific vendor or tool schema, while the corresponding output reflects its normalized representation in a canonical schema used by the framework. This design allows the agent to learn mappings and abstractions that generalize across formats, ensuring robust data transformation and alignment in support of higher-level situational-awareness modeling.

LLM development. The LLM development is treated as a machine translation task, where the LLM translates between data formats. The input includes the current and target format, with the LLM trained to output the data in the desired format. We hypothesize that vendor-provided format specifications will improve performance, generalizability, and handling of unseen formats. The framework accommodates multiple representations (raw text, summaries, structured metadata) for incorporating these specifications to determine which most effectively boosts performance. Examples of such specifications include tool guides, schema definitions, data dictionaries, and API documentation provided by vendors to describe the structure, semantics, and field mappings of their tools' formats.

Learning from feedback. Within this module, in-context learning (prompt-based approaches) and supervised fine-tuning are adopted to develop high-performing LLM agents. Normalized data generated by the agent support the construction of the graph model in Module 2, which feeds into Modules 3 and 4. To refine the LLM agent further, the framework incorporates feedback mechanisms into these subsequent modules, including strategies like RLHF, which uses human feedback to fine-tune models, aligning outputs with desired outcomes. In this context, RLHF iteratively improves the interpreter by leveraging cybersecurity analysts' evaluations of its normalized outputs' accuracy, relevance, and completeness.

The normalized data store does not simply preserve each source independently; instead, it maintains explicit links between related entities – such as alerts, logs, and threat-intelligence feeds – through shared identifiers and temporal correlations. This relational structure, represented in a JSON-based format, enables the framework to associate multiple sources with a single event context and supports the construction of the multi-graph representation used in Module 2.

The proposed framework complements, rather than replaces, existing SOC tools such as SIEM, SOAR, and threat-intelligence platforms. In Fig. 4, these existing systems can be viewed as data sources for Module 1, where configuration information is gathered and interpreted by the LLM-based data integration component, and Module 3, where alerts, logs, and network traffic information are collected for analysis. The framework's outputs, such as contextual risk assessments and remediation plans, can be fed back into orchestration tools for automated execution. This design supports incremental adoption within operational SOC environments without requiring substantial infrastructure changes.

5.2 Module 2: Context-Aware Cyber Situational Awareness

Module 2 provides advanced solutions for comprehensive and context-aware cyber situational awareness in SOCs and is focused on the following research question.

Research Questions. How can SOC achieve comprehensive, context-aware cyber situational awareness to enable more informed incident response decisions? How can the context be expanded beyond organizational boundaries to incorporate cyber threat intelligence from external sources and additional dimensions such as economic and geopolitical factors influencing malicious actors?

Traditional approaches focus on internal assets (*knowledge of us*) and partial knowledge of adversaries (*knowledge of them*). The framework extends situational awareness beyond organizational boundaries to include external factors — such as economic conditions, geopolitical tensions, and social dynamics — that influence the motivations and behaviors of malicious actors, ranging from cyber criminals to nation states. Diverse threat intelligence sources are aggregated and structured to create a more holistic view of the cyber threat landscape. By incorporating this multi-dimensional data, SOC can adapt their defense strategies to evolving threats. Contextual awareness can also improve both proactive and reactive cybersecurity measures, such as predicting attack vectors and prioritizing responses based on real-world implications. This module includes two sub-modules: Module 2.1 focuses on automatically analyzing diverse information streams, including news media, social media, and threat intelligence, to identify SOC-relevant contexts, assess risk, and identify potential targets [59]; Module 2.2 aims to integrate all available information into a comprehensive graphical model.

5.2.1 Module 2.1: LLM-based Real-Time Context Analysis. This module provides real-time context analysis capabilities for SOC by extending Topic Detection and Tracking (TDT) to integrate dynamic risk assessment. Originally a DARPA-sponsored natural language processing (NLP) task, TDT focuses on identifying and tracking emerging events in information streams [7]. We adopted TDT as it provides a well-established mechanism for identifying and tracking evolving topics over time, which aligns with the goal of maintaining situational awareness in dynamic cyber environments. While we focus on TDT as a representative approach, alternative methods such as temporal clustering, graph-based community detection, or embedding-based topic evolution could also be explored in future implementations, depending on data availability and performance requirements. The framework builds on TDT to create a near-real-time cybersecurity risk assessment framework that integrates LLMs and enables continuous monitoring of diverse data sources, such as news, social media, and threat intelligence feeds, facilitating the transition from event detection to actionable situational awareness tailored to an evolving, SOC-specific threat landscape. In traditional TDT, identifying and tracking emerging real-world events involves generating structured event representations using predefined ontologies with attributes and relations (e.g., trigger/action, location, or target) or clustering and linking text data to discover and monitor topics, without relying on explicit structuring [1, 10, 15, 62]. We can leverage the language understanding and reasoning capabilities of LLMs through Chain of Thought prompting strategies [61] to focus on ontology attributes relevant to risk assessment. These ontology attributes are initially derived from standard cybersecurity knowledge bases and domain expertise, and refined through iterative human validation supported by the embedded researcher. These attributes guide the LLM to consider the contextual information most relevant to SOC, consistent with recent evidence that ontology-grounded representations improve reasoning and task performance in domain-specific LLM applications [46, 60, 64]. For instance, when tracking a geopolitical event, the LLM can be directed to assess statements of hostility (trigger), geographic indicators (location), and the potential threat recipient (target). This attribute-driven focus helps the LLM prioritize relevant details, filter out extraneous information, and identify emergent risks.

The risk posed by an identified event to a specific SOC is framed as a classification problem. Inputs include risk assessment criteria, the event annotation ontology, a representation of the target, and a summary of the event. Outputs are categorical risk labels (e.g., no threat, low, medium, or high) indicating the perceived threat level. These labels inform Module 2.2's adaptive model for dynamic threat assessment. An effective classifier requires a comprehensive SOC representation that accounts for geopolitical, sociopolitical, and technical contexts and

generalizable criteria for distinguishing risk levels. It must distinguish between irrelevant content, relevant but non-threatening content, negative commentary lacking genuine risk, and content signaling legitimate threats. The Sony Pictures Entertainment cyber-attack 2 highlights the potential of real-time context analysis for enhancing SOC operations. The framework would process media coverage of the film and North Korea's hostile response, identifying these stories as both relevant and threatening. TDT captures North Korea's public condemnation of the film, particularly in state-run news outlets labeling it anti-propaganda [33]. By leveraging TDT ontology attributes, the LLM could focus on attributes related to the trigger/action (condemnation), sentiment (threatening or hostile), actor (North Korea), and target (Sony Pictures) in assessing relevancy and threat potential. Combining TDT annotation ontology and LLM language understanding yields a framework capable of identifying threats while filtering out irrelevant content, such as negative movie reviews or exaggerated social media posts. By prioritizing actionable insights, the system supports timely and informed responses. This LLM-based approach to TDT produces actionable event summaries and assigns risk labels for the Adaptive Data Modeling in Module 2.2. This enables SOC analysts to monitor emerging risks in real time and adapt defensive strategies in response to geopolitical tensions or other contextual factors, enhancing situational awareness and enabling timely responses. Validation is accomplished through retrospective analysis of prior incidents to uncover patterns and indicators relevant to threat prediction. Using data repositories like the Center for Strategic and International Studies' Significant Cyber Incidents list [22], representations for the applicable SOC are generated, utilizing historical media from tools like the Way Back Machine [9]. Analyzing this data refines the predictive capabilities of real-time systems, helping SOCs adapt to evolving global contexts. The real-time context analysis framework also includes feedback loops from subsequent tasks to refine the LLM-based TDT modeling, further enhancing the accuracy of context-aware incident response. To minimize the risks associated with retrospective validation, the framework evaluates model performance on time-bounded data streams to approximate real-time conditions. All incoming data are subject to proactive quality control through source whitelisting, human oversight, and validation prior to ingestion. Each dataset is version-controlled, and any subsequent corrections are applied through traceable, auditable updates to preserve transparency and reproducibility.

5.2.2 Module 2.2: Adaptive Data Modeling. The goal of Module 2.2 is to develop a dynamic and adaptive data modeling capability that can inform Module 3, supporting real-time risk assessment and effective prioritization of incident response efforts in SOCs. This submodule builds upon the data integration capabilities of Module 1 and the context-aware cyber situational awareness of Module 2.1, using a multi-graph model to capture system components, vulnerabilities, configurations, and external risk factors (e.g., geopolitical conditions). The novelty of this task lies in designing a scalable, continuously updated data model that adapts to the ever-evolving threat landscape and informs automated decision-making processes. The framework adopts a vulnerability-centric approach to data modeling, to enable prioritization of threat responses by focusing on exploitable weaknesses, improving real-time situational awareness and adaptive defenses. This capability builds upon our previous work in developing similar graphical data models [5, 52] and metrics to accurately assess the impact of vulnerability exploits in complex systems [4, 28] and reason about optimal metrics-driven mitigation strategies. Building upon this body of work, we design a multi-graph model, with subgraphs representing different classes of artifacts, including vulnerabilities, system components, configuration parameters, and external risk factors. These risk factors, identified via Module 2.1, include but are not limited to geopolitical and economic conditions, with nodes representing geopolitical events or economic sanctions, for example, that can influence the likelihood of attacks like nation-state-driven cyber espionage; compliance and regulatory requirements with nodes representing compliance mandates (e.g., GDPR, HIPAA) mapped to configuration or system component nodes to reflect how failure to meet regulatory changes impact system security; and other factors identified through various threat intelligence feeds (e.g., APT activity, emerging attack vectors) that can be used to dynamically adjust the vulnerability subgraph by adding newly discovered vulnerabilities, hypothesizing the presence of zero-day

vulnerabilities in seemingly secure but critical components, and adjusting severity scores accordingly. External factors in this context encompass, but are not limited to, traditional threat intelligence data. They also include operational and environmental dimensions, such as scheduled maintenance, ongoing campaigns, or changes in regulatory and geopolitical conditions, that shape how SOC analysts interpret and prioritize events. By integrating these broader sources of context, the framework aligns with existing SOC situational-awareness practices yet extends them by explicitly modeling non-threat contextual information as part of the reasoning process, thereby enriching the knowledge base that guides adaptive decision-making. By modeling these various risk factors and mapping them to other constructs in the different subgraphs, the multi-graph model becomes context-aware and capable of evolving in response to changing environmental conditions, such as geopolitical events or new regulatory frameworks.

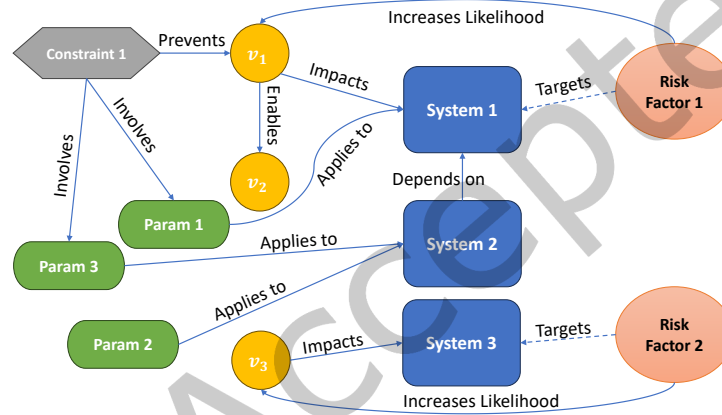


Fig. 5. Example of graphical model.

In our graphical data model, nodes represent key entities that capture different aspects of the system and its risk management environment, as shown in the in the notional example of Fig. 5. Vulnerability nodes (depicted in yellow) capture weaknesses or flaws in the system that can be exploited by threats, offering insights into potential security breaches and multi-step attacks [5, 39, 44] — enabled by the sequential exploitation of multiple dependent vulnerabilities — and their impact on system components. Component nodes (in blue) represent the physical or logical parts of a system at different levels of abstraction (individual components, subsystems, or systems), with the component subgraph capturing functional dependencies among the parts. Risk Factor nodes (in orange) define variables or conditions, identified through Module 2.1, that target specific systems and contribute to the likelihood or impact of security incidents, enabling a contextual assessment of threats and system exposure to external risks. Configuration nodes (in green) document the settings and parameters that govern the operation of the system, reflecting its current state and influencing its vulnerability and resilience to attacks. Constraints among configuration parameters (in gray) capture the role of configuration settings in mitigating vulnerability exposure. Together, these interconnected sets of nodes offer a rich representation of the system’s structure, weaknesses, and risk environment, providing a baseline knowledge for LLMs to reason about.

The multi-graph model is designed to scale efficiently with large and complex systems, allowing it to handle a growing number of system components, vulnerabilities, and external factors. Scalability is key to ensuring that the framework remains effective as organizations continue to expand their cybersecurity infrastructure and adopt new tools and technologies. To this aim, the model builds on our previous work on efficiently indexing patterns in graph data [6]. Building on our previous work on vulnerability metrics [4] and quantification of the

potential impact of cyber-attacks [28], we define families of metrics to quantify risk and guide the generation of courses of action that can provably reduce risk for the enterprise, paving the way for a novel Metrics-Augmented Generation (MAG). This system of metrics builds upon the two key vulnerability metrics defined in [4]: the exploitation likelihood $\rho(v)$ and the exposure factor $ef(v)$ of a vulnerability v , as defined by Equations 1 and 2 below.

$$\rho(v) = \frac{\prod_{x \in X_l^\uparrow} (1 - e^{-\alpha_x f_x(X(v))})}{\prod_{x \in X_l^\downarrow} e^{\alpha_x f_x(X(v))}} \quad (1)$$

$$ef(v) = \frac{\prod_{x \in X_e^\uparrow} (1 - e^{-\alpha_x f_x(X(v))})}{\prod_{x \in X_e^\downarrow} e^{\alpha_x f_x(X(v))}} \quad (2)$$

In these equations, $X_l^\uparrow$, $X_l^\downarrow$, $X_e^\uparrow$, and $X_e^\downarrow$ are sets of variables that influence the exploitation likelihood and exposure factors of vulnerabilities. Specifically, $X_l^\uparrow$ and $X_l^\downarrow$ are sets of variables that, respectively, contribute to increasing and decreasing the likelihood as their values increase. Similarly, $X_e^\uparrow$ and $X_e^\downarrow$ are sets of variables that, respectively, contribute to increasing and decreasing the exposure as their values increase.

Typical examples of variables in $X_l^\uparrow$ include, but are not limited to, a vulnerability's exploitability score, the time since details about the vulnerability were published, and the set of known exploits. In our work, we model the external risk factors discussed earlier as variables in $X_l^\uparrow$, allowing us great flexibility in modeling. Examples of variables in $X_l^\downarrow$ include, but are not limited to, the set of known Intrusion Detection System (IDS) rules associated with a vulnerability and the set of available vulnerability scanning plugins. Similarly, examples of variables in $X_e^\uparrow$ include a vulnerability's impact score, and examples of variables in $X_e^\downarrow$ include the set of deployed IDS rules associated with a vulnerability, which can help decrease the impact by detecting the onset of exploitation activity.

In these equations, each variable contributes to the overall likelihood or exposure factor as a multiplicative factor between 0 and 1 that is formulated to account for diminishing returns. Factors corresponding to a variable X in $X_l^\uparrow$ or $X_e^\uparrow$ are of the form $1 - e^{-\alpha_X \cdot f_X(X(v))}$ where α_X is a tunable parameter, $X(v)$ is the value of X for v , and f_X is a monotonically increasing function used to convert values of X to scalar values. Similarly, factors corresponding to a variable X in $X_l^\downarrow$ or $X_e^\downarrow$ are of the form $e^{-\alpha_X \cdot f_X(X(v))}$.

A key aspect of our approach is to model and quantify the role of external risk factors. In complex enterprise networks comprising multiple subsystems, each providing distinct services, various risk factors may impact systems differently. To capture these influences, we map each risk factor to the vulnerabilities existing on its target system or subsystem and incorporate these factors into the computation of vulnerability likelihood in Equation (1) as described earlier. In the notional example of Fig. 5, two risk factors respectively affect two different systems, so they can be modeled as variables influencing the likelihood of exposed vulnerabilities v_1 and v_3 respectively.

5.3 Module 3: LLM-Assisted Adaptive Decision-Making

Module 3 uses LLMs to dynamically assess and adapt to the ever-changing risk landscape, and it responds the following research questions.

Research Questions. How can SOC dynamically assess and adapt to the evolving risk landscape, and use LLM-based agents to support real-time risk assessment and prioritize incident response efforts? What are the most effective models for human-machine collaboration in SOC, to support decision-making and adapt to an ever-changing threat environment?

5.3.1 Module 3.1: Interrogating LLM for Context-aware Investigation. This submodule analyzes data from diverse feeds, including network traffic and alerts from various tools, inferences from larger contexts (Module 2.1), and risk indications calculated from the graphical model (Module 2.2), to suggest potential courses of actions both for investigation and remediation. Anthropological fieldwork provides an effective means of capturing tacit analyst knowledge to develop this capability. Embedded fieldworkers carry out participant observation [24, 53] and work as analysts in real-world SOC. Embedded fieldworkers play the roles of both the researcher and the analyst shown in Fig. 3. In carrying out the investigation, the analysts ask the LLM agent to perform subtasks such as looking for patterns of potential misuse in logs. The LLM is provided with all relevant contextual information from the various data feeds described above. It generates an initial output, including a rationale explaining its decision-making process. Analysts then interact with the LLM by refining its outputs through iterative natural language instructions, guiding the model to better align with their preferences. This refinement process could involve multiple rounds of revisions, where the LLM adjusts its reasoning and suggestions in response to user feedback. If the analysts are unable to achieve a satisfactory result through this interaction alone, they could directly edit the output to produce a revised version. These revised outputs, along with the original inputs, are then used to fine-tune the LLM, gradually aligning its responses with analyst preferences.

This approach builds on the concept of generative feedback [31], which emphasizes learning from natural language critiques to iteratively refine model outputs. Rather than relying on static supervision or RLHF, generative feedback enables LLMs to adapt dynamically through user interactions that provide fine-grained feedback on both the strengths and weaknesses of an initial response. By incorporating iterative refinement and direct human correction, the framework extends this paradigm to better support investigative workflows, allowing the LLM to capture subject matter expertise and institutional preferences over successive interactions. In the early stages, analysts may need to invest more effort in refining outputs, either by providing detailed prompting or by manually editing the model's responses. However, as the LLM incorporates this expertise through feedback-driven learning, the need for direct human intervention is expected to decrease. All interactions are recorded as op traces, as illustrated in Fig. 3. After sufficient improvement, the agent is piloted with production SOC analysts. At this stage, the fieldworkers only play the role of the researcher in Fig. 3, alongside SOC analysts, to observe how human analysts interact with the LLM agent. In addition to the op traces, the embedded researchers also record their observations in fieldnotes. These observations can reveal unspoken suppositions and other dimensions of tacit knowledge not captured in the conversations between human analysts and LLM agents. These tacit dimensions are useful in creating more effective learning signals for LLM agents to be improved through the learning process described below. The continuous evaluation and refinement aim to enhance the precision and effectiveness of human-machine co-teaming for investigating cyber incidents and making the right remediation decisions.

5.3.2 Module 3.2: Teaching LLM through SOC Experience. This submodule provides approaches that enable the LLM agents to learn from their experience in the SOC's investigative tasks. The learning signals generated through LLM agents' collaboration with human analysts come from two sources (Fig. 3): 1) recorded op traces, and 2) distilled explicit knowledge through qualitative analysis. To make the learning process sustainable, human analysts bear minimum burden in helping provide the learning signals. The conversations between the LLM agent and human analysts must be organic and for the sole purpose of moving forward with the investigation and finding the most appropriate remediations. In doing so some tacit knowledge is inevitably missed in the recorded op traces. This is where the embedded researcher can help. The researcher, trained in anthropological research

methods, records in the fieldnotes any observations that capture the context under which the interactions took place. The analysis of the fieldnotes provide additional learning signals from the explicated tacit knowledge through qualitative analysis. After the learning signals are formed, various learning strategies are applied as explained in Section 4.1. Different strategies are compared to understand their strengths and weaknesses and gain an understanding of when to utilize which strategies to yield the best outcome. After the LLM agent has learned substantial new knowledge, an updated version is deployed into operation, and the process repeats.

5.3.3 Considerations. A central element of adaptive decision-making is the process by which analysts and AI agents build mutual understanding and trust during multi-turn interactions. In the proposed framework, collaboration is not limited to information exchange but evolves through an iterative dialogue in which analysts probe, validate, and refine the agent’s reasoning. The LLM agent is designed to expose intermediate rationales and supporting evidence for its recommendations, allowing analysts to verify the soundness of its conclusions before acting. This transparency enables analysts to calibrate their trust in the agent over time, rewarding accurate, well-justified outputs and correcting misleading or incomplete ones. Although the current implementation focuses primarily on validating the feasibility of such interactions, developing systematic mechanisms for trust calibration remains an active area of ongoing research. On the other side, before analyst rationales are incorporated into model training, they undergo light preprocessing to remove sensitive information, normalize language and structure, and ensure compatibility with the input schema expected by the LLM. Specifically, to ensure that only reliable and contextually appropriate information informs model updates, all analyst–LLM interaction traces will initially be reviewed by the embedded researcher before being used as learning signals. This manual oversight serves two purposes: it filters out interactions containing factual errors or misleading reasoning, and it tags useful exchanges with qualitative codes capturing their relevance and outcome. Over time, this process will guide the creation of lightweight automated preprocessing pipelines, such as filtering based on confidence scores or rule-based quality checks, to support scalable supervision. Although these capabilities are still under development, establishing a structured oversight and preprocessing phase is essential to guarantee that model refinement is based on validated and trustworthy data.

5.4 Module 4: Explainable AI and Impact Quantification

Module 4 employs LLMs to embed explainable AI capabilities within SOC workflows, thereby improving transparency and trust in incident-response decisions and providing metrics to quantify the operational impact of AI adoption. The module combines rule-based reasoning with feature-attribution techniques, such as SHAP (SHapley Additive exPlanations), to make the decision process both interpretable and data-grounded. Rule-based reasoning exposes the logical sequence of inferences and contextual relationships that led to a recommendation, while SHAP-style attributions quantify the contribution of individual data features (e.g., configuration parameters, alert attributes, or vulnerability scores) to a specific output. Together, these complementary mechanisms allow analysts to trace how both contextual reasoning and data evidence informed the system’s conclusions, promoting trust and accountability in AI-assisted SOC operations. Thus, Module 4 addresses the following research question.

Research Questions. How can we leverage an LLM’s ability to articulate reasoning to achieve explainable AI integrated into SOC’s to enhance transparency, trust, and understanding of decision-making during incident response? How can the impact of adopting LLMs in SOC’s be measured, not only in terms of operational efficiency but also in incident detection and response effectiveness?

5.4.1 Module 4.1: LLM-based Report Generation. Once human analysts, assisted by LLM agents, have reached investigative conclusions and formulated remediation strategies, the SOC needs to produce reports for the relevant stakeholders who are impacted by the incident and/or remediations. The framework leverages LLM’s

capability to transform the analysis results from Module 3.1 into reports designed for various human stakeholders, emphasizing interpretability and transparency. The report provides a narrative explanation of the proposed actions, their rationale, and their alignment with risk metrics. It also includes contextual information, such as the expected impact, underlying assumptions, and data sources, enabling stakeholders to understand and trust the decision-making process. To help the AI develop potential courses of action, a template is developed by the research team with input from SOC team members. Each course of action presents SOC-specific details and clearly links the proposed steps to the particular security issue they address. Although full access to all intermediate data and reasoning steps would improve report completeness, the module can generate accurate summaries using contextual metadata and reasoning traces captured across prior modules, ensuring robustness even under partial data availability. Future extensions of this module will enable role-aware reporting, where LLMs adjust the level of technical detail and narrative style to match the needs of different stakeholders, from SOC analysts to organizational decision-makers.

5.4.2 Module 4.2: Plan Evaluation. In Module 4.2, the focus shifts to evaluating the proposed plan’s feasibility and effectiveness through a structured review by human analysts. This evaluation differs from the interactive refinement in Module 3.2 by incorporating a broader, post-generation assessment of the plan, including its alignment with organizational goals and long-term impact. Feedback from this evaluation refines both the analysis process in Module 3.1 and the report generation process in Module 4.1, ensuring iterative improvement and consistency with evolving security contexts.

5.4.3 Module 4.3: Plan Execution. Module 4.3 addresses the automation of the plan’s implementation, transforming the finalized plan from Module 4.1 into executable artifacts such as scripts, system configurations, or other machine-readable instructions. This task focuses on operationalizing the plan with minimal manual intervention while maintaining oversight mechanisms to ensure alignment with organizational policies. By bridging the gap between planning and action, Module 4.3 enables efficient and reliable execution of risk mitigation strategies, reinforcing the overall AI-assistance framework.

5.5 Practical Considerations and Limitations

While deploying local LLM infrastructure entails certain resource and governance considerations, embedding ethnographers within SOC teams does not represent a significant challenge. Our prior fieldwork demonstrates that such engagement can be accomplished efficiently through part-time or short-term immersion, leveraging existing collaborations with SOC partners. In fact, the overarching Research Question 2 (Section 4.2) explicitly calls for assessing how the resources invested in ethnographic study translate into measurable productivity gains for human-AI co-teaming. By contrast, hosting LLMs locally may require specialized compute resources and compliance processes to safeguard sensitive operational data. However, these requirements are aligned with current trends in SOC modernization and can be addressed through gradual, modular deployment — starting with small-scale prototypes integrated into existing SIEM or SOAR infrastructures.

6 Case Study and Preliminary Results

To begin assessing the feasibility and utility of the proposed multi-module architecture, we implemented prototypes of several components of the framework and conducted a proof-of-concept case study. As this is a primarily a position paper, the intent was not to deliver a comprehensive deployment, but rather to construct a thin yet functional pipeline that could test core capabilities across the first three modules and allow early validation of the human-machine co-teaming paradigm. Modules 1 and 2 were implemented as a minimal data integration and context-aware modeling layer, producing normalized, enriched, and structured data suitable for downstream reasoning. These data artifacts are suitable to provide context to Module 3, where an LLM agent was

integrated into a representative SOC workflow. Even though Modules 1, 2, and 3 are yet to be fully integrated into a production pipeline, the case study was designed to explore both the technical viability of such integration and its potential to reduce analyst burden in high-volume, cognitively repetitive tasks such as alert triage. In particular, the case study for Module 3 uses a real SOC's tasks and shows how our framework may be applied in practice.

6.1 Module 1 and 2

We developed a minimal yet functional instantiation of Module 1 (data-centric integration) and Module 2 (context-aware situational awareness). The goal was not a full deployment, but a thin data layer that normalizes heterogeneous inputs and exposes compact, context-aware signals suitable for the LLM agent defined for Module 3.

6.1.1 Module 1 Normalization and Enrichment. Raw inputs (representative assets, alerts, scanner findings) were first translated into a common JSON schema aligned with the STIX-inspired approach for Module 1 (key-value, LLM-legible fields). Specifically:

- We modeled organizational assets and their vulnerabilities as a directed heterogeneous graph: assets and vulnerabilities are distinct node types; edges model asset-to-asset connectivity and asset-vulnerability associations. CVE nodes are deduplicated to show shared exposure across assets.
- For each CVE, we fetched detailed descriptions from NVD 2.0 API (descriptions, timestamps, CVSS groups), preferring CVSS v3.1 over v3.0 or v2.0 and retaining base/impact/exploitability scores, vectors, and version tags.
- Relevant CWE information was fetched from the MITRE repository.

This normalization step directly produces normalized data that is consumed downstream.

6.1.2 Module 2 Adaptive Data Modeling (local vulnerability graph). From the normalized JSON, we populated a multi-graph tailored to this study's scope: vulnerabilities, components, and simple configuration aspects plus links that reflect functional and exposure dependencies. Fig. 6 shows a simplified example of such graph. While Section 5.2.2 lays out a richer, scalable model with risk-factor nodes (e.g., geopolitical, regulatory) and families of metrics, our case study used a simplified graph sufficient to prioritize alert investigation. Nonetheless, we enriched the model with vulnerability and weakness scores computed using the Mason Vulnerability Scoring Framework (MVSF) [29] to guide prioritization efforts.

Finally, the JSON file containing all the above information is converted into a vector database using LangChain, enabling seamless integration into a RAG-based pipeline for the AI agent.

6.2 Module 3

To begin validating our proposed human-machine co-teaming paradigm, we conducted an initial case study focused on a representative SOC task: alert triage. This case study is based on real SOC activities carried out at one of the co-authors' organizations. The goal of this case study was twofold.

- (1) Assess the feasibility of integrating a large language model (LLM)-based agent into analyst workflows.
- (2) Collect early evidence of productivity and knowledge-sharing benefits.

Alert triage was selected for this initial study because it is a core and frequently repeated activity in SOC operations. Analysts are assigned a number of alert tickets that must be investigated and resolved within a limited time frame, and new alerts are generated continuously. Depending on the scale and type of the organization, the number of alerts can range from a few dozen per day in smaller enterprise SOC's to tens of thousands per day in large organizations, making this task both operationally critical and highly repetitive. Fig. 7 illustrates the typical SOC alert triage workflow, highlighting how alerts are generated, enriched, and processed by analysts.

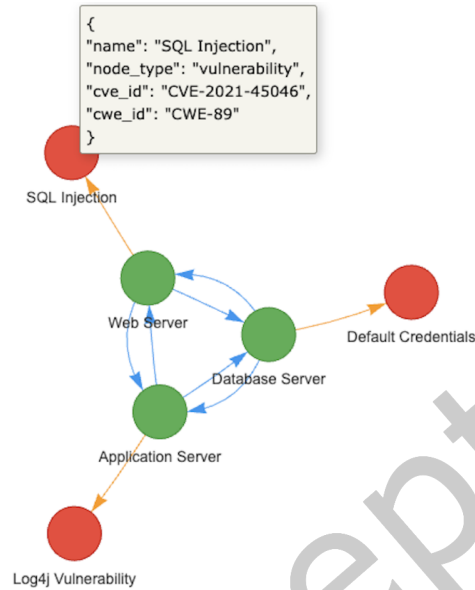


Fig. 6. Example of simple graph including system components, vulnerabilities, and their relationships.

The case study was conducted using a small set of real alert tickets obtained from a university Security Operations Center (SOC) in which the authors conducted an embedded observational study. The use of these tickets for this study was reviewed and authorized by the organization. Due to confidentiality agreements with the SOC, detailed descriptions of the underlying infrastructure and alert contents cannot be disclosed. For the purposes of this feasibility study, we evaluated the framework on ten representative SOC tickets covering common alert types encountered during routine analyst investigations. Because the objective of this case study is to demonstrate the feasibility of the proposed human-machine co-teaming workflow rather than to provide statistically generalizable results, the evaluation focuses on qualitative observations and preliminary performance indicators derived from these representative cases. Larger-scale empirical evaluations with operational SOC datasets remain an important direction for future work. We acknowledge that these confidentiality constraints limit direct reproducibility of the case study; accordingly, the reported results should be interpreted as an initial feasibility assessment rather than a reproducible benchmark.

6.2.1 Case Study Setup. We embedded researchers within a university SOC, where they worked as L1 SOC analysts to gain firsthand experience with the alert triage process. Based on their understanding of the workflow and the tools used in the SOC, they developed an LLM-based agent capable of processing alert tickets and interacting with external tools to enrich and validate the information contained in the alerts. The agent ultimately produces a structured report that can assist human analysts in making informed triage decisions.

The LLM agent was implemented in Python using the LangChain framework and configured as a ReAct-style agent capable of iterative reasoning and tool invocation. The agent was instantiated using the open-weight GPT-OSS model (approximately 20 billion parameters), deployed locally, and augmented with external tools to replicate key elements of a SOC analyst's workflow. These tools were integrated through API interfaces, enabling the agent to retrieve threat intelligence data and query SIEM records during the alert enrichment process.

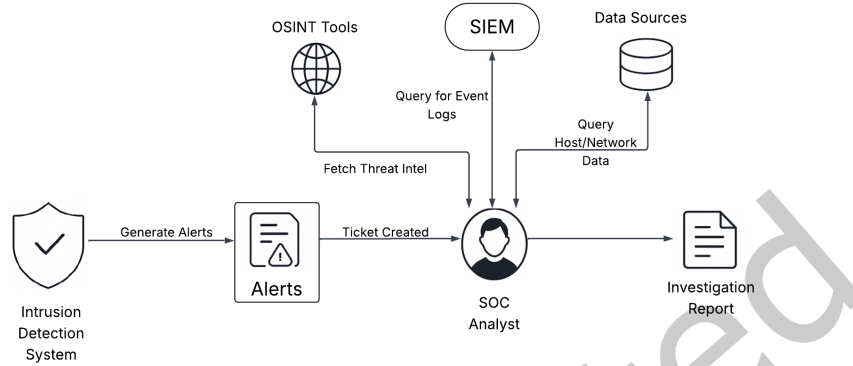


Fig. 7. SOC Analyst Workflow.

Specifically, VirusTotal was integrated for open-source intelligence (OSINT) queries, and Splunk served as the SIEM platform for retrieving network activity. The agent was tasked with processing alert tickets and producing the following outputs:

- (1) Natural language summaries of the alerts.
- (2) Enrichment with contextual threat intelligence from external sources.
- (3) Validation of alerts through evidence obtained from connected tools.
- (4) Suggested triage decisions, including initial severity assessments and recommended follow-up actions.

All outputs were logged for further analysis. In addition, we qualitatively evaluated the responses to assess whether the agent could reliably extract relevant indicators of compromise (IOCs) from each alert ticket.

6.2.2 Observations. Running the LLM agent on representative alert tickets revealed several notable patterns. First, the agent was able to quickly normalize heterogeneous alert data and produce natural language summaries that resembled the type of notes analysts would typically add to tickets. This suggests that the model can offload some of the repetitive documentation tasks that contribute to analyst fatigue.

Second, the enrichment process was generally effective in associating alerts with external threat intelligence. For instance, alerts involving suspicious domains were often cross-referenced with Virustotal results, providing contextual signals that could support prioritization. However, the quality of enrichment varied: in some cases, the model surfaced tangential or incomplete intelligence, indicating a need for tighter grounding in curated data sources.

Finally, the model demonstrated early signs of capturing tacit reasoning by highlighting contextual factors not explicitly coded into the alerts themselves. For example, certain IDS alerts were flagged as potentially low priority because the triggering activity matched patterns commonly associated with benign network scans. While these observations are promising, they also highlight the importance of a feedback mechanism to correct over-generalizations or misinterpretations. Fig. 8 shows how the AI agent worked through a typical SOC ticket step by step, using various tools along the way and finishing with a structured investigation report.

In this workflow, the LLM agent produces an initial enrichment of the alert, which is then reviewed and validated by the analyst. The analyst may confirm the relevance of the retrieved threat intelligence, discard hallucinated indicators, or refine the investigation by issuing additional queries. This interaction forms a human-machine co-teaming cycle in which the AI system accelerates information gathering while the analyst provides validation and corrective feedback. Although the current preliminary prototype does not implement automated model

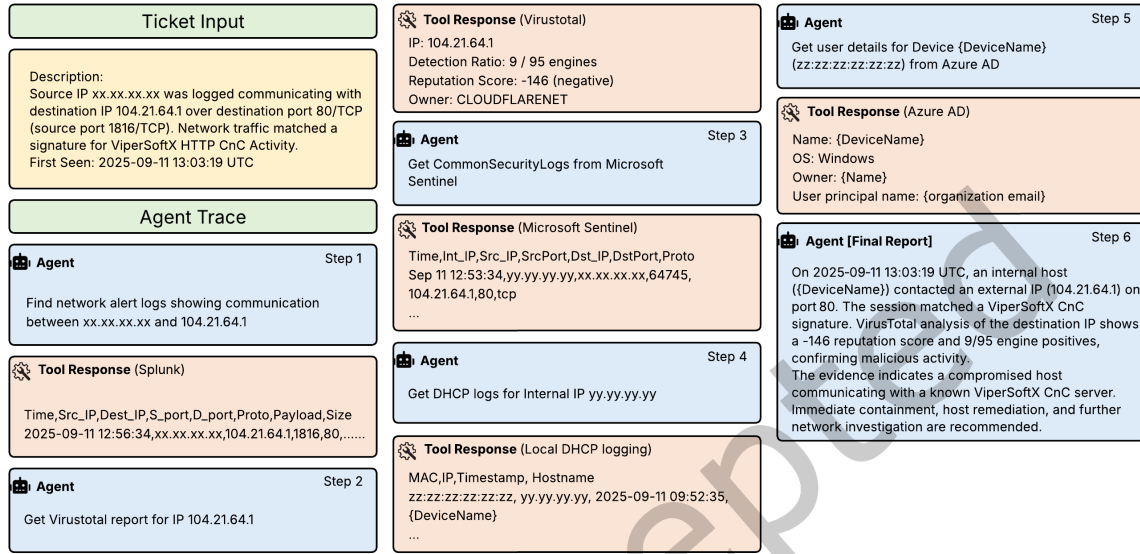


Fig. 8. Agent trace for a typical SOC ticket.

adaptation based on analyst feedback, this interaction pattern illustrates how analyst corrections guide subsequent investigation steps and provides a foundation for future adaptive learning mechanisms.

6.2.3 Preliminary Results. The evaluation of the agent's outputs focused on two dimensions: efficiency and quality, and was conducted on 10 real SOC alert tickets in the SOC the researchers were deployed in. Each alert ticket was processed twice: first using the LLM-assisted enrichment workflow and then through a manual investigation performed using the standard analyst workflow. The results were compared to assess whether the LLM agent successfully identified relevant indicators of compromise and contextual threat intelligence associated with the alert.

Efficiency. The model consistently produced summaries and enrichment in seconds, demonstrating clear potential for reducing the time spent on routine alert handling. Although this experiment did not include controlled timing measurements with human analysts, the results suggest that LLM-assisted enrichment could substantially reduce the time required to prepare initial triage summaries.

Quality. Across the ten evaluated tickets, the enrichment process successfully identified relevant threat intelligence in 80% of cases. In the remaining cases, the model generated hallucinated indicators or irrelevant associations that required analyst correction. Suggested triage decisions were directionally useful but would require human verification before being adopted in production. In the remaining 20% of cases, which were affected by hallucinations, the model occasionally generated spurious IP addresses, hashes, or domain names, which were then sent to the connected tools for enrichment, resulting in incorrect reports. These behaviors were observable because each tool invocation made by the LLM was logged and stored for analysis.

Overall, these preliminary results indicate that the LLM agent can shoulder some of the cognitive load associated with high-volume alert triage. At the same time, they underscore the importance of designing co-teaming mechanisms — such as retrieval grounding and iterative feedback loops — to ensure reliability and build analyst trust.

Table 1. Summary of open research challenges for AI-driven human-machine collaboration in SOC.

Challenge	Key Issue	Research Directions
Reliability and hallucination control	LLMs may generate unsupported conclusions in security-critical contexts.	Ground model reasoning in verifiable artifacts such as logs, telemetry, and threat intelligence feeds using retrieval-augmented generation, hybrid symbolic-LLM reasoning, and evidence attribution mechanisms.
Trust calibration and human oversight	Analysts must understand when AI outputs are reliable and when additional verification is required.	Design interaction models that expose confidence, uncertainty, and data provenance while enabling analysts to efficiently validate and refine AI-generated insights.
Learning from analyst feedback	Capturing tacit knowledge from analyst workflows and incorporating it into adaptive models remains challenging.	Investigate reinforcement learning from human feedback, incremental model updates, and structured mechanisms for collecting analyst feedback within SOC platforms.
Evaluation methodologies for co-teaming	Traditional ML metrics do not capture operational impact in SOC environments.	Develop evaluation frameworks that measure investigation time, analyst workload, and decision quality through controlled studies and SOC workflow simulations.
Privacy and deployment constraints	Operational SOC environments often require on-premises deployment and strict data governance.	Explore architectures for privacy-preserving deployment such as model distillation, secure enclaves, and federated learning to support LLM-assisted SOC tools.

7 Open Research Challenges and Future Directions

While recent advances in large language models create new opportunities for augmenting Security Operations Center (SOC) workflows, several fundamental research challenges remain before these capabilities can be reliably integrated into operational environments. The following challenges outline key open problems and corresponding research directions for realizing effective AI-driven human-machine collaboration in SOC. Table 1 provide a compact summary of these challenges and associated research directions.

Challenge 1: Reliability and hallucination control in security-critical reasoning. LLMs are known to occasionally generate hallucinated or unsupported conclusions, which can be particularly problematic in security contexts where decisions may affect incident response actions. SOC analysts rely on evidence-based reasoning derived from logs, alerts, and threat intelligence, whereas LLMs may generate plausible but incorrect interpretations when context is incomplete or ambiguous.

Future research should investigate methods for grounding LLM reasoning in verifiable security artifacts, such as structured logs, telemetry, and threat intelligence feeds. Possible directions include retrieval-augmented architectures, hybrid symbolic-LLM reasoning pipelines, and mechanisms for evidence attribution that allow analysts to trace generated conclusions back to specific data sources. Success in this area would involve AI systems that can provide traceable, evidence-backed reasoning that analysts can verify and audit during investigations.

Challenge 2: Trust calibration and human oversight in AI-assisted workflows. Effective human–machine collaboration requires appropriate calibration of trust between analysts and AI systems. Over-trust in automated outputs may lead analysts to overlook errors, while under-trust may cause analysts to ignore useful AI-generated insights. In SOC environments, where analysts operate under time pressure and cognitive load, designing interfaces that support effective trust calibration is a significant challenge.

Research directions include developing interaction models that explicitly communicate model confidence, uncertainty, and provenance of generated insights. Human-centered evaluation studies are also needed to understand how analysts interpret and act on AI-generated recommendations during incident investigations. Success would involve AI-assisted tools that allow analysts to efficiently validate and refine AI outputs while maintaining situational awareness and decision authority.

Challenge 3: Learning from analyst feedback at scale. A key motivation for human–machine co-teaming is the ability for AI systems to learn from the tacit knowledge embedded in analyst workflows. However, incorporating analyst feedback into adaptive systems raises challenges related to scalability, consistency, and robustness. In large SOC environments, thousands of alerts and tickets may be processed daily, and capturing meaningful feedback signals from analysts without disrupting their workflow remains difficult.

Future work should explore mechanisms for incorporating analyst feedback into adaptive models through techniques such as reinforcement learning from human feedback, incremental model adaptation, and structured feedback capture within SOC platforms. At the same time, safeguards are needed to prevent feedback loops that reinforce incorrect reasoning or introduce bias. Success would involve systems capable of continuously improving their performance while preserving reliability and stability in operational settings.

Challenge 4: Evaluation methodologies for human–AI co-teaming. Evaluating AI systems in SOC environments is challenging because success cannot be measured solely by traditional machine learning metrics such as accuracy or precision. Instead, the impact of AI assistance must be assessed in terms of operational outcomes, including analyst workload, investigation time, and decision quality.

Future research should develop evaluation methodologies that capture both technical performance and human–AI interaction outcomes. This may involve controlled experiments with analysts, simulation environments for SOC workflows, and longitudinal studies of AI-assisted operations. Success would involve evaluation frameworks that can rigorously measure the operational impact of AI-driven SOC augmentation.

Challenge 5: Privacy, deployment, and operational constraints. Many SOC environments operate under strict privacy, regulatory, and security constraints that limit the use of external cloud-based AI services. As a result, practical deployment of LLM-assisted SOC tools often requires on-premises or controlled-environment deployments, which introduces constraints related to model size, computational resources, and data governance.

Future work should investigate architectures for secure deployment of LLM-based SOC tools that support on-premises inference, controlled data access, and privacy-preserving learning mechanisms. Techniques such as model distillation, secure enclaves, and federated learning may play a role in enabling AI capabilities while respecting operational constraints. Success in this area would enable practical deployment of AI-assisted SOC systems in environments with strict security and compliance requirements.

Addressing these challenges will be essential to realizing the full potential of AI-driven human–machine collaboration in next-generation SOC environments. By advancing research along these directions, the community can move toward SOC platforms where AI systems and human analysts collaborate effectively to improve detection, investigation, and response to cyber threats.

8 Conclusions

This paper outlines a vision for AI-driven human–machine co-teaming in Security Operations Centers (SOCs), positioning LLM-powered AI agents as collaborative apprentices that learn from and work alongside human

analysts. We propose a paradigm in which LLM-based agents augment SOC workflows by internalizing tacit knowledge, supporting dynamic decision-making, and alleviating analyst burden without replacing human judgment. By reframing AI as a learning partner rather than a deterministic tool, we open new pathways for adaptive, context-aware SOC operations. We invite the research and practitioner community to explore this human-AI collaboration model, investigate its practical applications, and advance its development toward measurable improvements in SOC productivity and resilience.

While this work establishes the conceptual foundation for this paradigm, it also highlights several open challenges that will guide future research. Section 7 consolidates these challenges and outlines a research agenda for advancing human-machine collaboration in SOC environments. The current implementation focuses primarily on feasibility validation and a preliminary case study. Comprehensive evaluation of analyst-agent interaction quality, trust calibration, and long-term operational impact remain important directions for future work.

Advancing this vision will require larger-scale empirical studies across diverse SOC environments, deeper integration with existing security tools, and the development of rigorous evaluation methodologies for human-AI collaboration in operational settings. We hope this work stimulates further research toward practical, trustworthy, and scalable human-machine collaboration in next-generation SOC.

Acknowledgments

This work was partially supported by the National Science Foundation under award no. 2235102, the Office of Naval Research under award no. N00014-23-1-2538, and the National Security Agency under award no. H98230-22-1-0311. Related engagement supported by the Commonwealth Cyber Initiative under award no. N-4Q25-007 helped inform the authors' understanding of operational pain points in Security Operations Centers and concerns related to AI adoption for security operations. Any opinions, findings, conclusions, or recommendations expressed in this material are those of the authors and do not necessarily reflect the views of the funding agencies.

References

- [1] Z. Akhgari, M. Malekimajid, and H. Rahmani. 2022. Sem-TED: Semantic Twitter Event Detection and Adapting with News Stories. In *Proceedings of the International Conference on Web Research*. 61–69. doi:10.1109/ICWR54782.2022.9786234
- [2] M. N. Al-Mhiqani, R. Ahmad, Z. Z. Abidin, W. Yassin, A. Hassan, K. H. Abdulkareem, N. S. Ali, and Z. Yunus. 2020. A Review of Insider Threat Detection: Classification, Machine Learning Techniques, Datasets, Open Challenges, and Recommendations. *Applied Sciences* 10, 15 (2020), 5208. doi:10.3390/app10155208
- [3] B. A. Alahmadi, L. Axon, and I. Martinovic. 2022. 99% False Positives: A Qualitative Study of SOC Analysts' Perspectives on Security Alarms. In *Proceedings of the 31st USENIX Security Symposium (USENIX Security 2022)*. Boston, MA, USA, 2783–2800. <https://www.usenix.org/system/files/sec22-alahmadi.pdf>
- [4] M. Albanese, I. Iganibo, and O. Adebisi. 2023. A Framework for Designing Vulnerability Metrics. *Computers & Security* 132 (September 2023), 11 pages. doi:10.1016/j.cose.2023.103382
- [5] M. Albanese and S. Jajodia. 2018. A Graphical Model to Assess the Impact of Multi-Step Attacks. *The Journal of Defense Modeling and Simulation* 15, 1 (2018), 79–93. doi:10.1177/1548512917706043
- [6] M. Albanese, A. Pugliese, V. S. Subrahmanian, and O. Udrea. 2007. MAGIC: A Multi-Activity Graph Index for Activity Detection. In *Proceedings of the 2007 IEEE International Conference on Information Reuse and Integration (IRI 2007)*. Las Vegas, NV, USA, 267–272. doi:10.1109/IRI.2007.4296632
- [7] J. Allan, J. Carbonell, G. Doddington, J. Yamron, and Y. Yang. 1998. Topic Detection and Tracking Pilot Study: Final Report. In *Proceedings of the DARPA Broadcast News Transcription and Understanding Workshop*. 194–218. <https://ciir.cs.umass.edu/pubfiles/ir-137.pdf>
- [8] E. Almazrouei, H. Alobeidli, A. Alshamsi, A. Cappelli, R. Cojocar, M. Debbah, É. Goffinet, D. Hessel, J. Launay, and Q. Malartic. 2023. The Falcon Series of Open Language Models. *arXiv* (2023). doi:10.48550/arXiv.2311.16867
- [9] Internet Archive. [n. d.]. Wayback Machine. <http://web.archive.org/> [Accessed: Jan. 24, 2025].
- [10] M. Asgari-Chenaghlu, M.-R. Feizi-Derakhshi, L. Farzinvar, M.-A. Balafar, and C. Motamed. 2021. Topic Detection and Tracking Techniques on Twitter: A Systematic Review. *Complexity* 2021 (2021), Article ID 8833084. doi:10.1155/2021/8833084
- [11] S. Barnum. 2012. Standardizing Cyber Threat Intelligence Information with the Structured Threat Information Expression (STIX™). The MITRE Corporation. <https://www.mitre.org/sites/default/files/publications/stix.pdf>

- [12] M. Bayer, P. Kuehn, R. Shanehsaz, and C. Reuter. 2024. CySecBERT: A Domain-Adapted Language Model for the Cybersecurity Domain. *ACM Transactions on Privacy and Security* 27, 2 (2024), 1–20. doi:10.1145/3652594
- [13] H. R. Bernard, A. Wutich, and G. W. Ryan. 2016. *Analyzing Qualitative Data: Systematic Approaches*. SAGE Publications. <https://uk.sagepub.com/en-gb/eur/analyzing-qualitative-data/book240717>
- [14] S. Brin and L. Page. 1998. The Anatomy of a Large-Scale Hypertextual Web Search Engine. *Computer Networks and ISDN Systems* 30 (1998), 107–117. <https://www.sciencedirect.com/science/article/abs/pii/S016975529800110X>
- [15] R. Cantini and F. Marozzo. 2022. Topic Detection and Tracking in Social Media Platforms. In *Proceedings of the 1st International Conference on Pervasive Knowledge and Collective Intelligence on Web and Social Media*. Springer, 41–56. doi:10.1007/978-3-031-31469-8_3
- [16] Y. Chen, M. Cui, D. Wang, Y. Cao, P. Yang, B. Jiang, Z. Lu, and B. Liu. 2024. A Survey of Large Language Models for Cyber Threat Detection. *Computers & Security* (July 2024). doi:10.1016/j.cose.2024.104016
- [17] Y. Chen, Z. Ding, L. Alowain, X. Chen, and D. Wagner. 2023. DiverseVul: A New Vulnerable Source Code Dataset for Deep Learning-Based Vulnerability Detection. In *Proceedings of the 26th International Symposium on Research in Attacks, Intrusions, and Defenses (RAID 2023)*. 654–668. doi:10.1145/3607199.3607242
- [18] CrowdStrike. 2020. CrowdStrike’s Work with the Democratic National Committee: Setting the Record Straight. CrowdStrike Blog. <https://www.crowdstrike.com/blog/bears-midst-intrusion-democratic-national-committee/>
- [19] G. Deng, Y. Liu, V. Mayoral-Vilches, P. Liu, Y. Li, Y. Xu, T. Zhang, Y. Liu, M. Pinzger, and S. Rass. 2024. PentestGPT: Evaluating and Harnessing Large Language Models for Automated Penetration Testing. In *Proceedings of the 33rd USENIX Security Symposium (USENIX Security 24)*. Philadelphia, PA, USA, 847–864. <https://www.usenix.org/conference/usenixsecurity24/presentation/deng>
- [20] W. Fan, Y. Ding, L. Ning, S. Wang, H. Li, D. Yin, T.-S. Chua, and Q. Li. 2024. A Survey on RAG Meeting LLMs: Towards Retrieval-Augmented Large Language Models. In *Proceedings of the 30th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*. Barcelona, Spain, 6491–6501. doi:10.1145/3637528.3671470
- [21] M. D. Feters and E. B. Rubinstein. 2019. The 3 Cs of Content, Context, and Concepts: A Practical Approach to Recording Unstructured Field Observations. *The Annals of Family Medicine* 17, 6 (2019), 554–560. <https://pmc.ncbi.nlm.nih.gov/articles/PMC6846267/>
- [22] Center for Strategic and International Studies (CSIS). [n. d.]. Significant Cyber Incidents. Strategic Technologies Program. <https://www.csis.org/programs/strategic-technologies-program/significant-cyber-incidents> [Accessed: Jan. 24, 2025].
- [23] T.-M. Georgescu. 2020. Natural Language Processing Model for Automatic Analysis of Cybersecurity-Related Documents. *Symmetry* 12, 3 (2020), 354. doi:10.3390/sym12030354
- [24] W. Gunn and L. B. Løgstrop. 2014. Participant Observation, Anthropology Methodology and Design Anthropology Research Inquiry. *Arts and Humanities in Higher Education* 13, 4 (2014), 428–442. doi:10.1177/1474022214543874
- [25] A. Happe and J. Cito. 2023. Getting Pwn’d by AI: Penetration Testing with Large Language Models. In *Proceedings of the 31st ACM Joint European Software Engineering Conference and Symposium on the Foundations of Software Engineering (ESEC/FSE 2023)*. San Francisco, CA, USA, 2082–2086. doi:10.1145/3611643.3613083
- [26] I. Hasanov, S. Virtanen, A. Hakkala, and J. Isoaho. 2024. Application of Large Language Models in Cybersecurity: A Systematic Literature Review. *IEEE Access* 12 (2024), 176751–176778. doi:10.1109/access.2024.3505983
- [27] S. Hays and D. J. White. 2024. Employing LLMs for Incident Response Planning and Review. *arXiv* (March 2024). doi:10.48550/arxiv.2403.01271
- [28] I. Iganibo, M. Albanese, M. Mosko, E. Bier, and A. E. Brito. 2023. An Attack Volume Metric. *Security and Privacy* 6, 4 (July 2023), 23 pages. doi:10.1002/spy2.298
- [29] Ibifubara Iganibo, Massimiliano Albanese, Kaan Turkmen, Thomas R Campbell, and Marc Mosko. 2022. Mason Vulnerability Scoring Framework: A Customizable Framework for Scoring Common Vulnerabilities and Weaknesses.. In *SECRYPT*. 215–225.
- [30] A. Q. Jiang, A. Sablayrolles, A. Mensch, C. Bamford, D. S. Chaplot, D. de las Casas, F. Bressand, G. Lengyel, G. Lample, and L. Saulnier. 2023. Mistral 7B. *arXiv* (2023). doi:10.48550/arXiv.2310.06825
- [31] D. Jin, S. Mehri, D. Hazarika, A. Padmakumar, S. Lee, Y. Liu, and M. Namazifar. 2023. Data-Efficient Alignment of Large Language Models with Human Feedback Through Natural Language. Amazon Science. <https://www.amazon.science/publications/data-efficient-alignment-of-large-language-models-with-human-feedback-through-natural-language>
- [32] K. R. Jones, D. A. Brucker-Hahn, B. Fidler, and A. G. Bardas. 2023. Work-From-Home and COVID-19: Trajectories of Endpoint Security Management in a Security Operations Center. In *Proceedings of the 32nd USENIX Security Symposium (USENIX Security 2023)*. Anaheim, CA, USA, 2293–2310. <https://www.usenix.org/system/files/usenixsecurity23-jones.pdf>
- [33] S. Jong. 2015. The Interview as Anti-North Korean Propaganda. *NK News* (March 2015). <https://www.nknews.org/2015/03/the-interview-as-anti-north-korean-propaganda/>
- [34] W. Knight. 2023. Geoffrey Hinton tells us why he’s now scared of the tech he helped build. *MIT Technology Review* (May 2023). <https://www.technologyreview.com/2023/05/02/1072528/geoffrey-hinton-google-why-scared-ai/>
- [35] V. Kumar, D. Sinha, A. K. Das, S. C. Pandey, and R. T. Goswami. 2020. An Integrated Rule-Based Intrusion Detection System: Analysis on UNSW-NB15 Dataset and the Real-Time Online Dataset. *Cluster Computing* 23 (2020), 1397–1418. doi:10.1007/s10586-019-03008-x

- [36] D. Lende, A. Monkhouse, J. Ligatti, and X. Ou. 2023. Co-Creation in Secure Software Development: Applied Ethnography and the Interface of Software and Development. *Human Organization* 82, 1 (2023), 13–24. doi:10.17730/1938-3525-82.1.13
- [37] M. Levi, Y. Alluouche, D. Ohayon, and A. Puzanov. 2024. CyberPal.AI: Empowering LLMs with Expert-Driven Cybersecurity Instructions. *arXiv* (August 2024). doi:10.48550/arXiv.2408.09304
- [38] Z. Liu. 2024. A Review of Advancements and Applications of Pre-Trained Language Models in Cybersecurity. In *Proceedings of the International Symposium on Digital Forensics and Security*. San Antonio, TX, USA. doi:10.1109/isdfs60797.2024.10527236
- [39] Z. Liu, C. Wang, and S. Chen. 2008. Correlating Multi-Step Attack and Constructing Attack Scenarios Based on Attack Pattern Modeling. In *Proceedings of the 2008 International Conference on Information Security and Assurance (ISA 2008)*. 214–219. doi:10.1109/ISA.2008.11
- [40] Trend Micro. 2014. The Hack of Sony Pictures: What We Know and What You Need to Know. Trend Micro Security News. <https://www.trendmicro.com/vinfo/us/security/news/cyber-attacks/the-hack-of-sony-pictures-what-you-need-to-know>
- [41] Trend Micro. 2018. A Look into the Lazarus Group's Operations. Trend Micro Security News. <https://www.trendmicro.com/vinfo/us/security/news/cybercrime-and-digital-threats/a-look-into-the-lazarus-groups-operations>
- [42] B. Min, H. Ross, E. Sulem, A. P. Veyseh, T. H. Nguyen, O. Sainz, E. Agirre, I. Heintz, and D. Roth. 2023. Recent Advances in Natural Language Processing via Large Pre-Trained Language Models: A Survey. *Comput. Surveys* 5, 2 (September 2023). doi:10.1145/3605943
- [43] S. Mitra, S. Neupane, T. Chakraborty, S. Mittal, A. Piplai, M. Gaur, and S. Rahimi. 2024. LOCALINTEL: Generating Organizational Threat Intelligence from Global and Local Cyber Knowledge. *arXiv* (January 2024). doi:10.48550/arxiv.2401.10036
- [44] J. Navarro, A. Deruyver, and P. Parrend. 2018. A Systematic Survey on Multi-Step Attack Detection. *Computers & Security* 76 (July 2018), 214–249. doi:10.1016/j.cose.2018.03.001
- [45] I. Nonaka and H. Takeuchi. 1995. *The Knowledge-Creating Company: How Japanese Companies Create the Dynamics of Innovation*. Oxford University Press, New York. <https://academic.oup.com/book/52097>
- [46] Shirui Pan, Linhao Luo, Yufei Wang, Chen Chen, Jiapu Wang, and Xindong Wu. 2024. Unifying Large Language Models and Knowledge Graphs: A Roadmap. *IEEE Transactions on Knowledge and Data Engineering* (2024). doi:10.1109/TKDE.2024.3352100
- [47] M. Polanyi. 1966. The Logic of Tacit Inference. *Philosophy* 41, 155 (1966), 1–18. doi:10.1017/S0031819100066110
- [48] M. Polanyi. 1966. *The Tacit Dimension*. Doubleday & Company, Inc., New York.
- [49] K. Shashwat, F. Hahn, X. Ou, D. Goldgof, L. Hall, J. Ligatti, S. R. Rajagopalan, and A. Z. Tabari. 2024. A Preliminary Study on Using Large Language Models in Software Pentesting. In *Proceedings of the Workshop on SOC Operations and Construction (WOSOC 2024)*. San Diego, CA, USA, 1–7. <https://www.ndss-symposium.org/wp-content/uploads/wosoc2024-2-paper.pdf>
- [50] N. Shenoy and A. V. Mbaziira. 2024. An Extended Review: LLM Prompt Engineering in Cyber Defense. In *Proceedings of the International Conference on Electrical, Computer and Energy Technologies*. Sydney, Australia, 1–6. doi:10.1109/icecet61485.2024.10698605
- [51] G. Siracusano, D. Sanvito, R. Gonzalez, M. Srinivasan, S. Kamatchi, W. Takahashi, M. Kawakita, T. Kakumaru, and R. Bifulco. 2023. Time for aCTIon: Automated Analysis of Cyber Threat Intelligence in the Wild. *arXiv* (July 2023). doi:10.48550/arXiv.2307.10214
- [52] H. Soroush, M. Albanese, M. A. Mehrabadi, I. Iganibo, M. Mosko, J. Gao, D. Fritz, S. Rane, and E. Bier. 2020. SCIBORG: Secure Configurations for the IoT Based on Optimization and Reasoning on Graphs. In *Proceedings of the 8th IEEE Conference on Communications and Network Security (IEEE CNS 2020)*. 10 pages. doi:10.1109/CNS48642.2020.9162256
- [53] J. P. Spradley. 2016. *Participant Observation*. Waveland Press. <https://www.waveland.com/browse.php?t=689>
- [54] S. C. Sundaramurthy, A. G. Bardas, J. Case, X. Ou, M. Wesch, J. McHugh, and S. R. Rajagopalan. 2015. A Human Capital Model for Mitigating Security Analyst Burnout. In *Proceedings of the 11th Symposium on Usable Privacy and Security (SOUPS)*. Ottawa, Canada, 347–359. <https://www.usenix.org/system/files/conference/soups2015/soups15-paper-sundaramurthy.pdf>
- [55] S. C. Sundaramurthy, J. McHugh, X. Ou, S. R. Rajagopalan, and M. Wesch. 2014. An Anthropological Approach to Studying CSIRTs. *IEEE Security & Privacy* 12, 5 (Sept./Oct. 2014), 41–47. doi:10.1109/MSP.2014.84
- [56] S. C. Sundaramurthy, J. McHugh, X. Ou, M. Wesch, A. G. Bardas, and S. R. Rajagopalan. 2016. Turning Contradictions into Innovations or: How We Learned to Stop Whining and Improve Security Operations. In *Proceedings of the 12th Symposium on Usable Privacy and Security (SOUPS 2016)*. Denver, CO, USA, 237–251. <https://www.usenix.org/system/files/conference/soups2016/soups2016-paper-sundaramurthy.pdf>
- [57] D. R. Thomas. 2006. A General Inductive Approach for Analyzing Qualitative Evaluation Data. *American Journal of Evaluation* 27, 2 (2006), 237–246. doi:10.1177/1098214005283748
- [58] H. Touvron, T. Lavril, G. Izacard, X. Martinet, M.-A. Lachaux, T. Lacroix, B. Rozière, N. Goyal, E. Hambro, and F. Azhar. 2023. LLaMA: Open and Efficient Foundation Language Models. *arXiv* (2023). doi:10.48550/arXiv.2302.13971
- [59] P. Trubowitz and K. Watanabe. 2021. The Geopolitical Threat Index: A Text-Based Computational Approach to Identifying Foreign Threats. *International Studies Quarterly* 65, 3 (May 2021), 852–865. doi:10.1093/isq/sqab029
- [60] Ruize Wang, Duyu Tang, Nan Duan, Zhongyu Wei, Xuanjing Huang, Jianshu Ji, Guihong Cao, Daxin Jiang, and Ming Zhou. 2021. K-Adapter: Infusing Knowledge into Pre-Trained Models with Adapters. In *Findings of the Association for Computational Linguistics: ACL-IJCNLP 2021*. 1405–1418. doi:10.18653/v1/2021.findings-acl.121
- [61] J. Wei, X. Wang, D. Schuurmans, M. Bosma, B. Ichter, F. Xia, E. Chi, Q. V. Le, and D. Zhou. 2022. Chain-of-Thought Prompting Elicits Reasoning in Large Language Models. In *Advances in Neural Information Processing Systems*, Vol. 35. 24824–24837. doi:10.5555/3600270.

3602070

- [62] W. Xiang and B. Wang. 2019. A Survey of Event Extraction from Text. *IEEE Access* 7 (2019), 173111–173137. doi:10.1109/access.2019.2956831
- [63] J. Yang, H. Jin, R. Tang, and J. Tang. 2024. Harnessing the Power of LLMs in Practice: A Survey on ChatGPT and Beyond. *Commun. ACM* 67, 1 (January 2024), 70–79. doi:10.1145/3649506
- [64] Xiaoying Zhang, Baolin Peng, Kun Li, Jingyan Zhou, and Helen Meng. 2023. SGP-TOD: Building Task Bots Effortlessly via Schema-Guided LLM Prompting. In *Findings of the Association for Computational Linguistics: EMNLP 2023*. 13348–13369. doi:10.18653/v1/2023.findings-emnlp.891

Received 1 May 2025; revised 13 March 2026; accepted 6 May 2026